

SCE-014 — THREE-CATS REASONING, GAP WHY, AND STRUCTURAL COGNITIVE RUNTIME

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with chatGPT (OpenAI)

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Repository: <https://github.com/sizhet/Structural-Cognitive-Emergence-SCE>

Abstract

Structural Cognitive Emergence (SCE) proposes that cognition does not primarily emerge from statistical prediction, but from the growth, interaction, and adaptation of structural knowledge representations.

Previous SCE studies introduced:

- Differential Concept Formation
- Calling Graph Emergence
- Gap Detection and Why Generation
- Question Emergence
- Cross-Graph Gap Emergence

This work connects these components into a unified Structural Cognitive Runtime.

Using the classic "three cats" reasoning problem and the mechanism of Gap Why generation, we argue that reasoning, explanation, confidence formation, and cognitive growth can all be understood as operations over three fundamental structural layers:

- Differential Trees
- Calling Graphs
- Triggering Links

The resulting framework provides a structural explanation for few-shot reasoning, explainability, autonomous question generation, and future Autonomous Structural Intelligence (ASI) systems.

1. Introduction

One of the oldest puzzles in cognitive science is:

How can a child recognize a fourth cat after seeing only three cats?

Conventional statistical approaches typically explain this through pattern

generalization.

However, such explanations leave several questions unanswered:

- Why can humans generalize from extremely small samples?
- Why can humans explain their decisions?
- Why can humans generate new questions?
- Why can humans identify missing knowledge?

Structural Cognitive Emergence proposes a different answer.

The child is not merely memorizing observations.

The child is constructing and continuously refining structural representations.

2. Three Fundamental Structural Layers

Current SCE research repeatedly converges toward three core structural layers.

2.1 Calling Graph Layer

The Calling Graph represents:

- relations
- processes
- workflows
- causality
- dependencies

Examples:

Cat

- |— Eats Fish
- |— Chases Mouse
- |— Has Fur
- └— Lives With Humans

The Calling Graph answers:

What is connected?

2.2 Differential Tree Layer

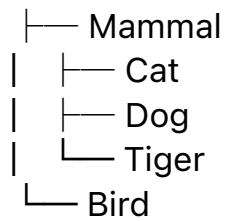
The Differential Tree represents:

- categories
- classifications
- naming
- assignments

- membership relations

Examples:

Animal



The Differential Tree answers:

What is it?

2.3 Triggering Links Layer

The Triggering Layer represents:

- threshold triggering
- state transitions
- decision triggers
- action initiation

Examples:

Hungry



Seek Food

Danger > Threshold



Escape

The Triggering Layer answers:

What happens next?

3. Three-Cats Reasoning

The classic cat problem can be reinterpreted structurally.

A child who has seen only three cats can still identify a fourth cat because

the child is not comparing images directly.
The child is performing structural reasoning.

Step 1: Differential Tree Localization

The observed object activates:

Animal
↓
Mammal
↓
Cat

This produces a candidate node.

Step 2: Calling Graph Expansion

The Cat node expands into connected knowledge.

Has Fur
Has Tail
Four Legs
Predatory Behavior
Domestic Animal

Step 3: Triggering Link Expansion

Additional behavioral evidence is activated.

If hungry → seek food

If threatened → escape

If sees prey → chase

Step 4: Scoring Tree Construction

Evidence is aggregated into a scoring structure.

Cat Score

+ Fur

- + Tail
- + Ear Shape
- + Eye Pattern
- + Behavior

The decision emerges from structural evidence accumulation.

4. Companion Node Reasoning

A particularly important mechanism is the generation of companion nodes. Instead of evaluating only:

Cat

the runtime simultaneously evaluates:

Cat

Dog

Tiger

Fox

Rabbit

Each candidate generates its own scoring tree.

Score(Cat)

Score(Dog)

Score(Tiger)

Score(Fox)

Score(Rabbit)

This creates decision depth and confidence estimation.

The resulting process is fundamentally different from a single matrix prediction.

5. Structural Confidence

Traditional statistical systems often produce confidence from probability distributions.

SCE proposes a different source.

Confidence emerges from:

Structural Depth

+
Cross-Evidence Consistency
+
Companion Competition

The more structural evidence supports a node while suppressing competing nodes, the stronger the confidence.
Thus confidence becomes a structural property rather than a purely statistical property.

6. Gap Why Generation

The same structural runtime explains Why generation.
Suppose we observe:

$A \rightarrow D$

but the expected Calling Graph contains:

$A \rightarrow B \rightarrow C \rightarrow D$

The runtime detects:

B ?
C ?

This missing region forms a structural gap.
The question:

Why did D happen?

is therefore equivalent to:

What structural nodes are missing?

7. Why as Gap Completion

SCE proposes:

Why Generation
=

Gap Detection
+
Gap Completion

This principle unifies:

- explanation generation
- scientific questioning
- autonomous coding
- hypothesis construction

under one structural mechanism.

8. Connection to Autonomous Coding

The connection is not accidental.

Autonomous Coding searches for:

Code Graph Gaps

Gap Why searches for:

Knowledge Graph Gaps

The underlying operation is identical.

Gap Detection
Gap Localization
Gap Expansion
Gap Completion

Therefore:

Why Generation

and

Autonomous Coding

can be viewed as different manifestations of the same structural process.

9. Toward Structural Cognitive Runtime

Combining previous SCE results yields a preliminary runtime model.

Differential Tree



Calling Graph



Triggering Links



Scoring Tree



Decision



Gap Why



Feedback Update

Or equivalently:

Locate



Expand



Trigger



Score



Decide



Explain



Update

This sequence forms a candidate Structural Cognitive Runtime.

10. Implications for ASI

Future Autonomous Structural Intelligence systems will require more than prediction.

They must be able to:

- identify structural gaps
- explain decisions
- generate questions
- evaluate competing hypotheses
- update structures through feedback

The runtime described here provides a possible foundation.

Without Gap Why capability, an AI may answer questions.

With Gap Why capability, an AI can discover questions.

This distinction may represent one of the critical transitions from statistical intelligence toward structural intelligence.

Conclusion

The Three-Cats Problem and Gap Why Generation are not isolated cognitive phenomena.

Both emerge from a common structural runtime operating over:

- Differential Trees
- Calling Graphs
- Triggering Links

Reasoning, confidence formation, explanation generation, autonomous questioning, and future ASI systems may therefore be understood as different manifestations of the same underlying structural process.

From this perspective, cognition is not primarily prediction.

Cognition is the continuous localization, expansion, evaluation, explanation, and restructuring of knowledge within a living structural system.